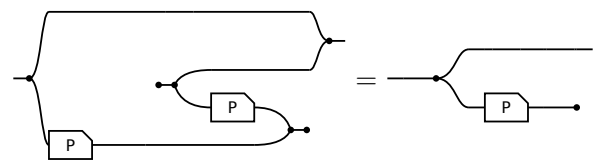
Relation Algebra

§5. Domain and range

(5a)

The **domain** `dom(R) ≜ 𝟙 ∩ (R R°)` and the **range** `ran(R) ≜ dom(R°)` .

(5b)



▷ lands the return leg back on the value ◁ handed out,

so R°▷ cuts to ⊸ — Frobenius

Running `R` and throwing the result away leaves only the fact that `R` could fire, and `ran(R)` is the same picture with the box mirrored. In `Rel` both steps are `{(a,a) : ∃b. a R b}`.

(5c)

1	`dom(R) ≜ 𝟙 ∩ (R R°)`
2	`Coreflexive(X) ⇒ dom(R) ⊑ X ⇔ R ⊑ X R`
3	`dom(R S) ⊑ dom(R)`
4	`dom(R ∩ S) = 𝟙 ∩ (S R°)`
5	`R` entire `Entire(R) ≜ dom(R) = 𝟙`
6	`R` simple `Simple(R) ≜ R° R ⊑ 𝟙`
7	`R` a map `Map(R) ≜ Entire(R) ∧ Simple(R)`

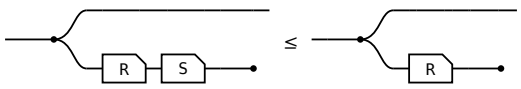
8	`Entire(R); Entire(S) ⇒ Entire(R S)`
	`Simple(f); Simple(g) ⇒ Simple(f g)`
	`Map(f); Map(g) ⇒ Map(f g)`

9	`Entire(R S) ⇒ Entire(R)`

§5.1. Sliding the discard

`dom(RS)⊑dom(R)`, and a single glyph for `dom` would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

(5.1a)



`S⊸⊑⊸`, the lax axiom for `⊸` in the first section — the discard slides back past `S`

Equality is `S` entire, which is the same picture read as `dom(R)=𝟙⇏R` entire.